



Chapter:- MOLE CONCEPT



Dr. Pankaj Sir
8787042016

Chemistry Planet

Chemistry Planet

Chemistry Planet

Chemistry Planet

CONCENTRATION TERMS

Q1- 8 g NaOH is dissolved in one litre of solution, its molarity is:

- (1) 0.8 M
- (2) 0.4 M
- (3) 0.2 M
- (4) 0.1 M

Q2- If 18 g of glucose is present in 1000 g of solvent, the solution is said to be:

- (1) 1 molar
- (2) 0.1 molar
- (3) 0.5 molar
- (4) 0.1 molal

Q3- The molarity of a solution of sodium chloride (mol wt. 58.5) in water containing 5.85 g of sodium chloride in 500 mL of solution is :-

- (1) 0.25
- (2) 2.0
- (3) 1.0
- (4) 0.2

Q4- For preparing 0.1 M solution of H_2SO_4 in one litre, we need H_2SO_4 :

- (1) 0.98 g
- (2) 4.9 g
- (3) 49.0 g
- (4) 9.8 g

Q5- Mole fraction of glycerine ($\text{C}_3\text{H}_5(\text{OH})_3$) in a solution of 36 g of water and 46 g of glycerine is:

- (1) 0.46
- (3) 0.20
- (2) 0.36
- (4) 0.40

Q6- 1000 g aqueous solution of CaCO_3 contains 10 g of calcium carbonate, concentration of the solution is:

- (1) 10 ppm
- (2) 100 ppm
- (3) 1000 ppm
- (4) 10,000 ppm

Q7- What is the normality of 1M H_3PO_4 solution?

- (1) 0.5 N
- (2) 1.0 N
- (3) 2.0 N
- (4) 3.0 N

Q8- The molarity of 0.2 N Na_2CO_3 solution will be:

- (1) 0.05 M
- (2) 0.2 M
- (3) 0.1 M
- (4) 0.4 M

Q9- Normality of 0.3 M phosphorous acid is:-

- (1) 0.15
- (2) 0.6
- (3) 0.9
- (4) 0.1

Q10- The molarity of pure water is:

- (1) 100 M
- (2) 55.5 M
- (3) 50 M
- (4) 18M

Q11- Molarity of 720 g of pure water-

- (1) 40M
- (2) 4M
- (3) 55.5M
- (4) Can't be determined

Q12- Equal weight of NaCl and KCl are dissolved separately in equal volumes of solutions then molarity of the two solutions will be -

- (1) Equal
- (2) That of NaCl will be less than that of KCl
- (3) That of NaCl will be more than that of KCl
- (4) That of NaCl will be half of that of KCl

Q13- The mole fraction of oxygen in a mixture of 7g of nitrogen and 8g of oxygen is:

- (1) $\frac{8}{15}$
- (2) 0.5
- (3) 0.25
- (4) 1.0



Chapter:- MOLE CONCEPT



Dr. Pankaj Sir
8787042016

Chemistry Planet

Chemistry Planet

Chemistry Planet

Chemistry Planet

Q14- In a solution of 7.8 g benzene (C_6H_6) and 46.0g toluene ($C_6H_5CH_3$) the mole fraction of benzene is:-

- (1) $\frac{1}{6}$
- (2) $\frac{1}{5}$
- (3) $\frac{1}{2}$
- (4) $\frac{1}{3}$

Q15- An X molal solution of a compound in benzene has mole fraction of solute equal to 0.2. The value of X is:-

- (1) 14
- (2) 3.2
- (3) 1.4
- (4) 2

Q16- Mole fraction of ethanol in ethanol water mixture is 0.25. Hence percentage concentration of ethanol by weight of mixture is

- (1) 25%
- (2) 75%
- (3) 46%
- (4) 54%

Q17- A 500 g tooth paste sample has 0.02 g fluoride concentration. What is the concentration of fluorine in terms of ppm level:-

- (1) 250
- (2) 40
- (3) 400
- (4) 1000

Q18- The normality of 10 mL of a "20 V" H_2O_2 solution is

- (1) 1.79
- (2) 3.58
- (3) 60.86
- (4) 6.086

Q19- H_2O_2 solution used for hair bleaching is sold as a solution of approximately 5.0 g H_2O_2 per 100 ml. of the solution. The molecular mass of

H_2O_2 is 34. The molarity of this solution is approximately

- (1) 0.15 M
- (2) 1.5 M
- (3) 3.0 M
- (4) 3.4 M

Q20- Normality of 10% (W/V) H_2SO_4 solution is nearly

- (1) 0.1
- (2) 0.2
- (3) 0.5
- (4) 2

Q21- What volume of 0.1 N HNO_3 solution can be prepared from 6.3 g of HNO_3 ?

- (1) 1 litre
- (2) 2 litre
- (3) 0.5 litre
- (4) 5 litre

Q22- A molal solution is one that contains one mole of a solute in

- (1) 1000 g of the solvent
- (2) one litre of the solution
- (3) one litre of the solvent
- (4) 22.4 litres of the solution.

Q23- Which of the following statement is true:

- (a) Molarity is the no. of moles of solute dissolved per litre of solvent.
 - (b) The molarity and normality of a solution of sodium carbonate are same.
 - (c) Molality (m) of a solution is defined as the number of moles of solute dissolved is 1000 gm of solution
 - (d) The ratio of mole fraction of solute and solvent is in the ratio of there respective moles.
- (1) a & c
 - (3) b & c
 - (2) a & d
 - (4) Only d



Chapter:- MOLE CONCEPT Dr. Pankaj Sir 8787042016

Chemistry Planet Chemistry Planet Chemistry Planet Chemistry Planet

Q24- Two bottles of A and B contains 1M and 1m aqueous solution ($d=1\text{gml.}$) of sulphuric acid respectively

- (1) A is more concentrated than B
- (2) B is more concentrated than A
- (3) Concentration of A = concentration of B
- (4) It is not possible to compare the concentration

Q25- Molar concentration of a solution in water is:

- (1) Always equal to normality
- (2) More than molality of the solution
- (3) Equal to molality of the solution
- (4) Less than the molality of the solution

Q26- The molarity of 98% H_2SO_4 ($d = 1.8 \text{ g mL}^{-1}$) by wt. is

- (1) 6 M
- (2) 18 M
- (3) 10 M
- (4) 4 M

SOLUBILITY (HENRY'S LAW)

Q27- Henry's law constant for dissolution of CH_3 in benzene at 298 K is $2 \times 10^5 \text{ mm of Hg}$. Then solubility of CH_3 in benzene at 298 K under 760 mm of Hg is:

- (1) 1.2×10^{-5}
- (2) 3.8×10^{-3}
- (3) 4×10^{-7}
- (4) 1×10^{-2}

Q28- Which of the following gas does not obey Henry's law ?

- (1) NH_3
- (2) H_2
- (3) O_2
- (4) He

VAPOUR PRESSURE (LIQUID-LIQUID MIXTURE)

Q29- 1 mol of heptane (V. P. 92 mm of Hg) was mixed with 4 mol of octane (V. P. = 31 mm of Hg). The vapour pressure of resulting ideal solution is:

- (1) 46.2 mm of Hg

(2) 40.0 mm of Hg

(3) 43.2 mm of Hg

(4) 38.4 mm of Hg

Q30- At 88 °C benzene has a vapour pressure of 900 torr and toluene has a vapour pressure of 360 torr. What is the mole fraction of benzene in the mixture with toluene that will boil at 88 °C at 1 atm. pressure, benzene toluene form an ideal solution:

- (1) 0.416
- (2) 0.588
- (3) 0.688
- (4) 0.740

Q31- P_A^O and P_B^O are 108 and 36 torr respectively. What will be the mole fraction of A in vapour phase if B has mole fraction in solution 0.5.

- (1) 0.25
- (2) 0.75
- (3) 0.60
- (4) 0.35

Q32- What is correct relation between mole fraction in vapour phase (Y_A) of A in terms of X_A If mole fraction in solution of A is (X_A). (If P_A^O vapour pressure of A in pure state)

- (1) $(1 - X_A) P_A^O$
- (2) $\frac{X_A}{1 - X_A} P_A^O$
- (3) $\frac{1 - X_A}{X_A} P_A^O$
- (4) $\frac{P_A^O X_A}{P_s} P_A^O$

IDEAL AND NON-IDEAL SOLUTIONS

Q33- Which condition is not satisfied by an ideal solution

- (1) $\Delta H_{\text{mixing}} = 0$
- (2) $\Delta V_{\text{mixing}} = 0$
- (3) $\Delta S_{\text{mixing}} = 0$
- (4) Obedience of Raoult's law

Q34- Among the following, that does not form an ideal solution is:

- (1) C_6H_6 and $\text{C}_6\text{H}_5\text{CH}_3$



Chapter:- MOLE CONCEPT



Dr. Pankaj Sir
8787042016

Chemistry Planet

Chemistry Planet

Chemistry Planet

Chemistry Planet

(2) C_2H_5Cl and C_6H_5OH

(3) C_6H_5Cl and C_6H_5Br

(4) C_2H_5Br and C_2H_5I

Q35- An azeotropic mixture of two liquids has b.p. lower than either of them when it

(1) shows a (+ve) deviation from Raoult's law

(2) shows no deviation from Raoult's law

(3) shows (+ve) deviation from Henry's law

(4) shows (-ve) deviation from Henry's law

COLLIGATIVE PROPERTIES

Q36- Which is not a colligative property?

(1) Osmotic pressure

(2) Relative lowering in vapour pressure

(3) Depression in freezing point

(4) Refractive index

Q37- The lowering of vapour pressure of a solvent by addition of a non-volatile solute to it is directly proportional to

(1) The strength of the solution

(2) The nature of the solute in the solution

(3) The atmospheric pressure

(4) All

Q38- The relative lowering of vapour pressure is equal to the mole fraction of the nonvolatile solute, This statement was given by

(1) Raoult

(2) Henry

(3) Joule

(4) Dalton

Q39- Which one of the following is the incorrect form of Raoult's law

$$(1) \frac{P_S}{P^O} = \frac{N}{n+N}$$

$$(2) \frac{P^O}{P^O - P_S} = 1 + \frac{N}{n}$$

$$(3) \frac{P^O - P_S}{P_S} = \frac{n}{n+N}$$

$$(4) \frac{P_S}{P^O - P_S} = \frac{N}{n}$$

Q40- The vapour pressure of a solution having solid as solute and liquid as solvent is:

(1) Directly proportional to mole fraction of the solvent

(2) Inversely proportional to mole fraction of the solvent

(3) Directly proportional to mole fraction of the solute

(4) Inversely proportional to mole fraction of the solute

Q41- P_o and P_s the vapour pressure of solvent and its solution respectively. N_1 and N_2 are the mole fraction of solvent and solute respectively then

$$(1) P_S = \frac{P_o}{N_2}$$

$$(2) P_S - P_o = P_o N_2$$

$$(3) P_S = P_o N_2$$

$$(4) \frac{P_o - P_S}{P_S} = \frac{N_1}{N_1 + N_2}$$

Q42- One mol of non volatile solute is dissolved in two mol of water. The vapour pressure of the solution relative to that of water is

(1) $2/3$

(2) $1/3$

(3) $1/2$

(4) $3/2$

Q43- The vapour pressure of a dilute aqueous solution of Glucose is 750 mm of mercury at 373 K. The mole fraction of solute is

(1) $1/10$

(2) $1/7.6$

(3) $1/35$

(4) $1/76$

Q44- The vapour pressure of water at room temperature is 23.8 mm of Hg. The vapour pressure of an aqueous solution of sucrose with mole fraction 0.1 is equal to

(1) 23.9 mm Hg

(3) 21.42 mm Hg

(2) 24.2 mm Hg

(4) 31.44 mm Hg



Chapter:- MOLE CONCEPT



Dr. Pankaj Sir
8787042016

Chemistry Planet

Chemistry Planet

Chemistry Planet

Chemistry Planet

Q45- The vapour pressure of pure A is 10 torr and at the same temperature when 1 g of B is dissolved in 20 gm of A, its vapour pressure is reduced to 9.0 torr. If the molecular mass of A is 200 amu, then the molecular mass of B is

- (1) 100 amu
- (3) 75 amu
- (2) 90 amu
- (4) 120 amu

Q46- The boiling point of C_6H_6 , CH_3OH , $C_6H_5NH_2$ and $C_6H_5NO_2$ are $80^\circ C$, $65^\circ C$, $184^\circ C$ and $212^\circ C$ respectively. Which will show highest vapour pressure at room temperature

- (1) C_6H_6
- (3) $C_6H_5NH_2$
- (2) CH_3OH
- (4) $C_6H_5NO_2$

Q47- If Raoult's law is obeyed, the vapour pressure of the solvent in a solution is directly proportional to

- (1) Mole fraction of the solvent
- (2) Mole fraction of the solute
- (3) Mole fraction of the solvent and solute
- (4) The volume of the solution

Q48- The vapour pressure of a pure liquid 'A' is 70 torr at $27^\circ C$ forms an ideal solution with another liquid B. The mole fraction of B is 0.2 and total vapour pressure of the solution is 84 torr at $27^\circ C$ The vapour pressure of pure liquid B at $27^\circ C$ is-

- (1) 14
- (2) 56
- (3) 140
- (4) 70

Q49- The boiling point of an aqueous solution of a non volatile solute is $100.15^\circ C$. What is the freezing point of an aqueous solution obtained by diluting the above solution with an equal volume of water? The values of K_b and K_f for water are 0.512 and 1.86 K

molality⁻¹:

- (1) - $0.544^\circ C$
- (3) - $0.272^\circ C$
- (2) - $0.512^\circ C$
- (4) - $1.86^\circ C$

Q50- The molal elevation constant is the ratio of the elevation in B.P. to:

- (1) Molarity
- (2) Molality
- (3) Mole fraction of solute
- (4) Mole fraction of solvent

Q51- Elevation in boiling point was $0.52^\circ C$ when 6 g of a compound x was dissolved in 100 g of water. Molecular weight of x is ($K=5.2 K mol^{-1} 100g H_2O$)

- (1) 120
- (2) 60
- (3) 100
- (4) 342

Q52- An aqueous solution containing 1g of urea boils at $100.25^\circ C$. The aqueous solution containing 3g of glucose in the same volume will boil at -

- (1) $100.75^\circ C$
- (2) $100.5^\circ C$
- (3) $100^\circ C$
- (4) $100.25^\circ C$

Q53- Pure benzene freezes at $5.45^\circ C$ at a certain place but a 0.374 m solution of tetrachloroethane in benzene freezes at $3.55^\circ C$ The K_f for benzene is-

- (1) $5.08 K kg mol^{-1}$
- (2) $508 K kg mol^{-1}$
- (3) $0.508 K kg mol^{-1}$
- (4) $50.8^\circ C K kg mol^{-1}$

Q54- An aqueous solution freezes at $0.186^\circ C$ ($K_1 = 1.86 K kg mol^{-1}$ $K_b = 0.512 K kg mol$). What is the elevation in boiling point?

- (1) 0.186
- (2) 0.512



Chapter:- MOLE CONCEPT Dr. Pankaj Sir 8787042016

Chemistry Planet

Chemistry Planet

Chemistry Planet

Chemistry Planet

- (3) 0.512/1.86
(4) 0.0512
- Q55- Camphor is used as solvent to determine the molecular weight of nonvolatile solute by Rast method because for camphor
(1) Molal depression constant is high
(2) Melting point is high
(3) Being cheap
(4) All
- Q56- In the depression of freezing point experiment, it is found that:-
(1) The vapour pressure of solution is more than of pure H_2O .
(2) The vapour pressure of solution is less than that of pure solute
(3) Only solute molecules solidify at of freezing point
(4) Only solvent molecules solidify at freezing point
- Q57- Molal depression constant of water is $1.86 K kg mol^{-1}$. 0.02 mol of urea dissolved in 100 g of water will produce a depression in freezing point of:
(1) $0.186^\circ C$
(2) $0.372^\circ C$
(3) $1.86^\circ C$
(4) $3.72^\circ C$
- Q58- What would be the freezing point of aqueous solution containing 18 g of $C_6H_{12}O_6$ in 1000 g of water? ($K_f = 1.86 K molality^{-1}$)
(1) $-0.186^\circ C$
(2) $-0.372^\circ C$
(3) $-0.54^\circ C$
(4) $-0.72^\circ C$
- Q59- A solution of 1.25 g of a non-electrolyte in 20 g of water freezes at 271.94 K. If $K_f = 1.86 K$ then the molecular wt. of the solute is:
molality $^{-1}$
(1) $207.8 g mol^{-1}$
(3) $209.6 g mol^{-1}$
- (2) $179.79 g mol^{-1}$
(4) $96.01 g mol^{-1}$
- Q60- If a thin slice of sugar beet is placed i laced in c solution of NaCl then in concentrated
(1) Sugar beet will lose water from its cells
(2) Sugar beet will absorb water from solution
(3) Sugar beet will neither absorb nor lose water
(4) Sugar beet will dissolve in solution
- Q61- In osmosis phenomenon net flow of:
(1) Solvent molecules occurs from higher concentration to lower concentration
(2) Solvent molecules occurs from lower concentration to higher concentration
(3) Solute molecules occurs from higher C concentration to lower concentration
(4) Solute molecules occurs from lower concentration to higher concentration
- Q62- At constant temperature the osmotic pressure of a solution is:
(1) Directly proportional to the concentration
(2) Inversely proportional to the concentration
(3) Directly proportional to the square of concentration
(4) Directly proportional to the square root of concentration
- Q63- Which inorganic precipitate acts as semipermeable membrane?
(1) Calcium sulphate
(2) Barium oxalate
(3) Nickel phosphate
(4) Copper ferrocyanide
- Q64- If 0.1 M solution of glucose and 0.1 M urea solution are placed on two sides of a semipermeable membrane to equal heights, then it will be correct to say that:
(1) There will be not net movement across the membrane
(2) Glucose will flow towards urea solution



Chapter:- MOLE CONCEPT



Dr. Pankaj Sir
8787042016

Chemistry Planet

Chemistry Planet

Chemistry Planet

Chemistry Planet

- (3) Urea will flow towards glucose solution
(4) Water will flow from urea solution towards glucose solution.
- Q65- The best colligative property used for the determination of molecular masses of polymers is:
- (1) Relative lowering in vapour pressure
 - (2) Osmotic pressure
 - (3) Elevation in boiling point
 - (4) Depression in freezing point
- Q66- If mole fraction of the solvent in solution decreases then:
- (1) Vapour pressure of solution increases
 - (2) B. P. decreases
 - (3) Osmotic pressure increases
 - (4) All are correct
- Q67- The osmotic pressure of a solution increases if:
- (1) Temperature is lowered
 - (2) Volume is increased
 - (3) Number of solute molecules is increased
 - (4) None
- Q68- Which of the following solutions at the same temperature will be isotonic:
- (1) 3.42 g of cane sugar in one litre water and 0.18 g of glucose in one litre water.
 - (2) 3.42 g of cane sugar in one litre water and 0.18 g of glucose in 0.1 litre water.
 - (3) 3.42 g of cane sugar in one litre water and 0.585g of NaCl in one litre water.
 - (4) 3.42 g of cane sugar in one litre water and 1.17 g of NaCl in one litre water.
- Q69- If density of 2 molal sucrose solution is 1.4 g mL^{-1} at 25°C , find osmotic pressure.
- (1) 4.06 atm
 - (2) 2 atm
 - (3) 40.6 atm
 - (4) 3.4 atm
- Q70- If total concentration of dissolved particles inside red blood cells is 0.3 M (approximate)

- and the membrane surrounding the cell is semipermeable. Find osmotic pressure inside the cell
- (1) 7.34 atm
 - (2) 1.78 atm
 - (3) 2.34 atm
 - (4) 0.74 atm
- Q71- Equal volume of 0.1 M urea and 0.1 M glucose are mixed. The mixture will have :-
- (1) Lower osmotic pressure
 - (2) Same osmotic pressure
 - (3) Higher osmotic pressure
 - (4) None of these
- Q72- Osmotic pressure of a solution (density is 1 g mL^{-1}) containing 3 g of glucose (molecular weight = 180) in 60 g of water at 15°C is:
- (1) 0.34 atm
 - (2) 0.65 atm
 - (3) 6.25 atm
 - (4) 5.57 atm
- Q73- Osmotic pressure of a sugar solution at 24°C is 2.5 atmosphere. The concentration of the solution in mole per litre is:
- (1) 10.25
 - (2) 1.025
 - (3) 1025
 - (4) 0.1025
- Q74- A solution containing 4 g of a non volatile organic solute per 100 ml was found to have an osmotic pressure equal to 500 cm of mercury at 27°C . The molecular weight of solute is:
- (1) 14.97
 - (2) 149.7
 - (3) 1697
 - (4) 1.497
- Q75- A solution containing 8.6 g urea in one litre was found to be isotonic with 0.5% (wt./vol)



Chapter:- MOLE CONCEPT



Dr. Pankaj Sir
8787042016

Chemistry Planet

Chemistry Planet

Chemistry Planet

Chemistry Planet

solution of an organic, non volatile solute. The molecular weight of later is:

- (1) 348.9
- (2) 34.89
- (3) 3489
- (4) 861.2

Q76- If a 6.84% (w/V) solution of cane-sugar (mol. wt. 342) is isotonic with 1.52% (w/V) solution of thiocarbamide, then the molecular weight of thiocarbamide is:

- (1) 152
- (2) 76
- (3) 60
- (4) 180

Q77- A solution containing 500 g of a protein per litre is isotonic with a solution containing 3.42 g of sucrose per litre. The molecular mass of protein is

- (1) 5
- (2) 146
- (3) 34200
- (4) 50000

Q78- The osmotic pressure of blood is 7.65 atm. at 310 K. an aqueous solution of Glucose that will be isotonic with blood is w/N:

- (1) 5.41%
- (3) 3.5%
- (2) 54.1%
- (4) 4.53%

ABNORMAL COLLIGATIVE PROPERTIES

Q79- Equimolal solutions of A and B show depression in freezing point in the ratio of 2: 1. A remains in normal state in solution. B will be in state in solution:

- (1) Normal
- (2) Associated
- (3) Hydrolysed
- (4) Dissociated

Q80- van't Hoff factor is:

- (1) Less than one in case of dissociation

(2) More than one in case of association

(3) Always less than one

(4) Less than one in case of association

Q81- The vant Hoff factor (i) for a dilute solution of $K_3[Fe(CN)_6]$ is:

- (1) 10
- (2) 4
- (3) 5
- (4) 0.25

Q82- The experimental molecular weight of an electrolyte will always be less than its calculated value because the value of vant Hoff factor, i is:

- (1) Less than 1
- (2) Greater than 1
- (3) One
- (4) Zero

Q83- The vant Hoff factor (i) for a dilute aqueous solution of Glucose is:

- (1) Zero
- (2) 1.0
- (3) 1.5
- (4) 2.0

Q84- 0.585% NaCl solution at 27°C has osmotic pressure.

- (1) 2.49 atm
- (2) 5 atm
- (3) 1.2 atm
- (4) 3.8 atm

Q85- The substance A when dissolved in solvent B shows the molecular mass corresponding to A_3 The vant Hoff's factor will be -

- (1) 1
- (2) 2
- (3) 3
- (4) 1/3

Q86- The ratio of the value of any colligative property for KCl solution to that for sugar solution is nearly time:

- (1) 1



Chapter:- MOLE CONCEPT



Dr. Pankaj Sir
8787042016

Chemistry Planet

Chemistry Planet

Chemistry Planet

Chemistry Planet

- (2) 0.5
(3) 2
(4) 2.5
- Q87- The lowering of vapour pressure of 0.1M aqueous solutions of NaCl, CuSO₄, and K₂SO₄ are:
(1) All equal
(2) In the ratio of 1: 1: 1.5
(3) In the ratio of 3: 2:1
(4) In the ratio of 1.5: 1: 2.5
- Q88- The freezing point of 1 molal NaCl solution assuming NaCl to be 100% dissociated in water is: ($K_f = 1.86\text{K Molality}^{-1}$)
(1) - 1.86 °C
(2) - 3.72 °C
(3) +1.86 °C
(4) +3.72 °C
- Q89- The molal elevation constant of water is 0.51. The boiling point of 0.1 molal aqueous NaCl solution is nearly:
(1) 100.05 °C
(2) 100.1 °C
(3) 100.2° C
(4) 101.0° C
- Q90- The molal elevation constant of water is 0.52K molality⁻¹. The boiling point of 1.0 molal aqueous KCl solution (assuming complete dissociation of KCl), should be:
(1) 100.52°C
(2) 101.04°C
(3) 99.48°C
(4) 98.96°C
- Q91- Solute A is ternary electrolyte and solute B is non-electrolyte. If 0.1 M solution of solute B produces an osmotic pressure of 2P, then 0.05M solution of A at the same temperature will produce an osmotic pressure equal to:
(1) P
(2) 1.5 P
(3) 2 P

- (4) 3 P
- Q92- Phenol associates in benzene as 92
$$\text{C}_2\text{H}_5\text{OH} \rightleftharpoons \frac{1}{2} (\text{C}_6\text{H}_5\text{OH})_2$$

If degree of association of phenol is 40%. van't Hoff factor i is
(1) 1
(2) 0.8
(3) 1.4
(4) 0.6
- Q93- The values of observed and calculated molecular weight of calcium nitrate are respectively 65.6 and 164. The degree of dissociation of calcium nitrate will be:
(1) 25%
(2) 50%
(3) 75%
(4) 60%
- Q94- A 0.004M solution of Na₂SO₄ is isotonic with a 0.010M solution of glucose at the 25°C temperature. The apparent degree of dissociation of Na₂SO₄ is
(1) 25%
(2) 50%
(3) 75%
(4) 85%
- Q95- A 5.8% (wt./vol.) NaCl solution will exert an osmotic pressure closest to which one of the following:
(1) 5.8% (wt./vol) sucrose solution
(2) 5.8% (wt./vol) glucose solution
(3) 2 M sucrose solution
(4) 1 M glucose solution
- Q96- Which salt shows maximum osmotic pressure in its 1M solution:
(1) AgNO₃
(2) Na₂SO₄
(3) (NH₄)₃PO₄
(4) MgCl₂
- Q97- Which solution will exert highest osmotic pressure?



Chapter:- MOLE CONCEPT



Dr. Pankaj Sir
8787042016

Chemistry Planet

Chemistry Planet

Chemistry Planet

Chemistry Planet

- (1) 1M glucose solution
(2) 1M urea solution
(3) 1M Alum solution
(4) 1M NaCl solution
- Q98- Which is the correct relation between osmotic pressure of 0.1M NaCl solution and 0.1M Na_2SO_4 solution?
(1) The osmotic pressure of Na_2SO_4 is less than NaCl solution
(2) The osmotic pressure Na_2SO_4 is more than NaCl solution
(3) Both have same osmotic pressure
(4) None of the above
- Q99- Which one of the following solutions will have highest osmotic pressure? (Assume that all the salts are equally dissociated)
(1) 0.1M $\text{Al}_2(\text{SO}_4)_3$
(2) 0.1M BaCl_2
(3) 0.1 M Na_2SO_4
(4) The solution obtained by mixing equal volumes of (2) and (3)
- Q100- The following solutions have equal concentrations. Which one will show minimum osmotic pressure?
(1) BaCl_2
(2) AgNO_3
(3) Na_2SO_4
(4) $(\text{NH}_4)_3\text{PO}_4$
- Q101- The osmotic pressure of equimolar solutions of BaCl_2 , NaCl, and glucose will be in the order:
(1) Glucose > NaCl > BaCl_2
(2) BaCl_2 > NaCl > Glucose
(3) NaCl > BaCl_2 > Glucose
(4) NaCl > Glucose > BaCl_2
- Q102- Which one of the following pairs of solutions will be expected to be isotonic under the same temperature?
(1) 0.1M urea and 0.1 M NaCl
(2) 0.1M urea and 0.2 M MgCl_2

- (3) 0.1M NaCl and 0.1M Na_2SO_4
(4) 0.1M $\text{Ca}(\text{NO}_3)_2$ and 0.1M Na_2SO_4
- Q103- Two solutions of KNO_3 and CH_3COOH are prepared separately. Molarity of both is 0.1M and osmotic pressures are P_1 and P_2 respectively. The correct relationship between the osmotic pressures is
(1) $P_2 > P_1$
(2) $P_1 = P_2$
(3) $P_1 > P_2$
(4) $\frac{P_1}{P_1+P_2} = \frac{P_2}{P_1+P_2}$
- Q104- The correct relationship between the boiling points of very dilute solutions of AlCl_3 (T_1) and CaCl_2 (T_2), having the same molar concentration is:
(1) $T_1 = T_2$
(2) $T_1 > T_2$
(3) $T_2 = T_1$
(4) $T_2 \geq T_1$
- Q105- Which of the following 0.1 M aqueous solutions will have the lowest freezing point:
(1) Potassium Sulphate
(2) Sodium Chloride
(3) Urea
(4) Glucose
- Q106- Which aqueous solution has minimum freezing point?
(1) 0.01 M NaCl
(2) 0.005 M $\text{C}_2\text{H}_2\text{O}$
(3) 0.005 M MgI ,
(4) 0.005 M MgSO_4 ,
- Q107- If α is the degree of dissociation of $\text{K}[\text{Fe}(\text{CN})_6]$, J. then abnormal mass of complex in the solution will be:-
(1) $M_{\text{normal}}(1+2\alpha)^{-1}$
(2) $M_{\text{normal}}(1+3\alpha)^{-1}$
(3) $M_{\text{normal}}(1+\alpha)^{-1}$
(4) $M_{\text{normal}}(1+4\alpha)^{-1}$
- Q108- Which solution will have least vapour pressure:



Chapter:- MOLE CONCEPT



Dr. Pankaj Sir
8787042016

Chemistry Planet

Chemistry Planet

Chemistry Planet

Chemistry Planet

- (1) 0.1 M BaCl,
(2) 0.1 M urea
(3) 0.1 M Na SO
(4) 0.1 M Na₃PO
- Q109- Which has maximum freezing point:
(1) 1 molar of NaCl solution
(2) 1 molar of KCl solution
(3) 1 molar of CaCl₂ solution
(4) 1 molar of urea solution
- Q110- The freezing point of 1% calcium nitrate will be: aqueous solution of
(1) 0°C
(2) Above 0°C
(3) 1°C
(4) Below 0°C
- Q111- The following aqueous solution in the correct order of decreasing freezing point is
(1) 0.2M BaCl, 0.2M KCl, 0.1M Na₂SO₄
(2) 0.2M KCl, 0.1M Na₂SO₄, 0.2M BaCl₂
(3) 0.1M Na₂SO₄, 0.2M KCl, 0.2M BaCl₂
(4) 0.1M Na₂SO₄, 0.2M BaCl₂, 0.2M KCl
- Q112- Which of the following solutions will have highest boiling point?
(1) 1% Glucose in water
(2) 1% Sucrose in water
(3) 1% NaCl in water
(4) 1% Urea in water
- Q113- The freezing point of will be highest for: equimolal aqueous solution
(1) C₆H₅NH₃Cl
(2) Ca(NO₃)₂
(3) La(NO₃)₃
(4) C₆H₁₂O₆ (Glucose)
- Q114- When mercuric Iodide is added to the aqueous solution of potassium iodide?
(1) The boiling point does not change
(2) Freezing point is raised
(3) The freezing point is lowered
(4) Freezing point does not change

- Q115- The molecular weight of benzoic acid in benzene as determined by depression in freezing point method corresponds to
(1) Ionisation of benzoic acid
(2) Dimerization of benzoic acid
(3) Trimerization of benzoic acid
(4) Solvation of benzoic acid.